

NDN Threat Detection — Federated & Continual Learning

M.Tech research · Named Data Networking · Technical product context

Case summary

M.Tech research on detecting collusive and non-collusive Interest Flooding Attacks in Named Data Networking using federated and continual learning.

NDN	networking
FL	federated learning
CL	continual learning
ML	detection

Problem

Interest Flooding Attacks can consume network and forwarding resources. The research focuses on distinguishing collusive and non-collusive attack behavior.

Approach

The work explores distributed model training through federated learning and adaptation over time through continual learning.

Evaluation mindset

The research context includes model behavior, distributed data, drift and detection performance—constraints that matter when AI systems are deployed in changing environments.

Product relevance

Demonstrates technical depth for AI and network products where reliability, detection, deployment constraints and model adaptation influence product decisions.

AI tools & workflow

AI-assisted research and coding support can accelerate experimentation, but the simulation design, dataset construction, model evaluation and conclusions remain engineering/research work rather than AI-generated claims.

Proof of work

This case-study artifact is included directly in the portfolio so recruiters can inspect the underlying product thinking rather than only read a short project card.